

Counterfactual Analysis in Knowledge Graph Based Recommender System

by

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ABSTRACT

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Contents

1	Introduction	7
1.1	Background and Motivation	7
1.2	Research Problem and Objective	11
1.3	Methodological Approach	12
1.4	Expected Contributions and Outcomes	12
1.5	Structure of the Thesis	13
2	Literature Review	15
2.1	Explainable AI In Knowledge Graph Recommender System	15
2.1.1	Path-based Modeling	15
2.1.2	Explainability through Symbolic and Logical Reasoning	16
2.2	Counterfactual Reasoning in XAI	17
2.3	Counterfactual Methods in Different Types of Recommender Systems	20
2.4	Application of Counterfactuals for Fairness and Bias Mitigation	22
2.4.1	Leveraging Knowledge Graphs for Fair Recommendations	23
3	Research Design and Methodology	24
3.1	Recommendation Engine/System Utilization	25
3.1.1	Data and Implementation	25
3.1.2	Knowledge Graph Composition	25
3.1.3	Path-Based Recommendation Mechanism	26
3.2	Counterfactual Analysis	27
3.3	Extraction of Relevant Information	27
4	Results	34
4.1	Case Study	34
4.1.1	Initial Recommendations	34

4.1.2	Counterfactual Analysis for User 0’s Recommendations	36
4.1.3	Key Marketing Findings and Enhancing Marketing Strategies . . .	43
4.2	Ablation Study: Impact of Community Detection and Outlier Removal . .	44
4.2.1	Experimental Setup	44
4.2.2	Results Overview	45
4.2.3	Comparison Across Additional Users	45
4.2.4	Discussion	46
4.3	Limitations and Challenges	46
5	Conclusion	48
5.1	Summary of Key Findings	48
5.2	Contributions to Explainable AI in Recommender Systems	49
5.3	Marketing and Business Implications	49
5.4	Future Research Directions	49
	References	51
	Appendix	54

Chapter 1

Introduction

1.1 Background and Motivation

The rapid growth of the Internet as a hub for digital and commercial interactions, which consequently generates vast amounts of data, has underscored the essential role of recommender systems in today's technology landscape. These systems, which are increasingly important to companies from media giants like Netflix to e-commerce leaders such as Amazon, make use of user feedback to create personalized experiences. This feedback varies from explicit ratings to implicit behavioral data, such as browsing and purchasing histories, and serves as a critical foundation for these systems. By analyzing past interactions between users and products, recommender systems can predict future preferences and use advanced algorithms to detect patterns in user behavior. These capabilities improve user satisfaction by aligning recommendations with individual tastes and enhance business strategies by promoting customer engagement and retention. As user data grows more complex, these systems evolve by integrating sophisticated models that consider content attributes, user specifications, and contextual information, thus increasing the accuracy and relevance of the recommendations offered (Aggarwal, 2016).

Despite their effectiveness in predicting user preferences and enhancing service personalization, traditional recommender systems, like many other AI systems, often operate as opaque entities. This opacity arises from their complex computational mechanisms that frequently use deep learning methods, known for their 'black-box' nature, where the internal reasoning processes remain hidden from both users and developers. This lack of transparency poses significant challenges, especially from a business perspective, where understanding and trust in these systems are of high value.

For businesses, the drive to make recommender systems explainable is motivated by several key factors. Transparency builds trust; when users understand how recommendations are generated, they are more likely to trust and accept these suggestions, thereby increasing user satisfaction and loyalty. This is particularly important in sectors like e-commerce or content streaming, where the accuracy of recommendations directly affects user engagement and retention.

Furthermore, explainability aids in complying with regulatory standards, such as the European Union's General Data Protection Regulation (GDPR), which requires that users must understand the decisions made by automated systems that affect them. This regulation is critical for businesses to ensure that their AI systems can provide clear explanations for their outputs.

Additionally, from a business innovation standpoint, explainability enables the refinement and optimization of recommender systems. By understanding the decision-making processes, developers can identify and address biases or errors, leading to more accurate and fair recommendations. This not only improves the system's performance but also increases its fairness and accountability, aligning with ethical AI practices.

The development of explainable AI (XAI) seeks to clarify AI operations, allowing both users and developers to understand, trust, and effectively manage AI outputs. Recent trends toward this field highlight the growing recognition of the need for AI systems, including recommender systems, to be both effective and understandable. As such, the business case for explainable recommender systems is strong, promising to enhance user experience and comply with regulations but also to foster technological advancement and ethical AI practices (Vultureanu-Albisi & Badica, 2021)

Explainable recommender systems can be understood from different angles. One way for their classification is their approach to explainability.

Types of Explainable Recommender Systems

- **Intrinsic Explainability:** This approach involves developing interpretable models where the decision-making process is transparent. As a result, it's easier to provide natural explanations for the decisions made by the model.
- **Model-Agnostic Approach:** Also known as the post-hoc explanation approach, this allows the decision mechanism to remain a black box. Instead of making the decision process itself clear, it focuses on developing a separate explanation model that generates explanations after a decision has been made.

The underlying philosophy of these approaches aligns closely with theories in human cognitive psychology. Sometimes, we make decisions through careful, rational reasoning and can clearly explain why we made those decisions. At other times, we make decisions first and then rationalize them afterward to support or justify our actions.

There are multiple approaches to achieving intrinsic explainability in recommender systems. These include factorization-based, topic modeling, graph-based, specialized deep learning methods, knowledge-based, and rule mining approaches. Knowledge graphs, in particular, play a critical role in advancing the field of Explainable Artificial Intelligence (XAI) by providing a structured and semantically rich framework that enhances the interpretability of complex AI models. As Tiddi and Schlobach (2022) describe, knowledge graphs are typically structured as directed, edge-labeled graphs that describe entities and their relationships, often organized within an ontological schema covering a variety of topics. This organization reflects human cognitive processes of understanding and reasoning, making it a transparent layer in AI systems where the decisions of otherwise opaque models can be traced and understood through clear, logical pathways. The primary value of knowledge graphs in explainability rises from their ability to connect AI outputs with comprehensible and verifiable pieces of information, offering deeper insights into the reasoning behind AI decisions. Therefore, knowledge graphs act as entities that both enhance the trustworthiness and clarity of AI systems and also play a significant role in bridging the gap between human understanding and machine reasoning, necessary for the broader adoption and ethical deployment of AI technologies (Tiddi & Schlobach, 2022).

Within the field of knowledge graph-based recommender systems, two main approaches stand out: Embedding-based approaches make use of the rich semantic information available within knowledge graphs to improve the representation of users and items. These methods may focus on creating item-specific graphs that enrich item representations with attributes from the knowledge graph or develop user-item graphs that include both entities and use the defined relationships and attributes to predict user preferences.

Conversely, path-based methods leverage the connectivity patterns within the graph, using paths to define and measure the similarity and interaction context between users and items. These methods often employ predefined meta-paths to capture semantic similarities and refine representations, thereby enhancing the recommendation process (Q. Guo et al., 2020).

While these explanations significantly improve transparency, they often do not fully satisfy the human cognitive need for deeper understanding. Human cognition naturally

operates by imagining alternative conditions and exploring various possible scenarios. By considering the outcomes of these hypothetical situations, individuals gain confidence in their grasp of a concept. Users frequently question how recommendations might change if product attributes were altered, such as brand changes or descriptors like "animal cruelty-free" or "plant-based."

While these explanations significantly improve transparency, they often do not fully satisfy the human cognitive need for deeper understanding. Human cognition naturally operates by imagining alternative conditions and exploring various possible scenarios. By considering the outcomes of these hypothetical situations, individuals gain confidence in their grasp of a concept. From a business perspective, understanding that the recommender system suggested Product X due to Attribute A is valuable, but knowing how recommendations might change if the product had different attributes—such as an altered brand, or descriptors like "animal cruelty-free" or "plant-based" can provide businesses with high-value strategic insights into how these changes could influence user perceptions and preferences. This study aims to extend the level of explainability of path-based knowledge-graph-based recommendation through counterfactual analysis. Counterfactuals represent hypothetical alternatives to actual events, offering insights into what could have happened under different circumstances. These constructs are crucial in various domains of artificial intelligence, especially in enhancing the interpretability and trustworthiness of AI systems, a fundamental aspect of explainable AI. Counterfactuals serve as a critical tool in XAI by providing "what if" analyses that make clear the decision-making processes of AI models, showing how alternative inputs might alter an AI systems output and making these systems more transparent and understandable.

The application of counterfactuals in XAI are present in both intrinsic and post-hoc approaches. Intrinsic methods integrate counterfactual reasoning as part of the AI model's fundamental design, directly influencing its operational framework. Conversely, post-hoc techniques generate counterfactual explanations after the model has made a decision. By simulating different scenarios, counterfactuals provide insights into the specific conditions under which the outcomes of AI decisions might change, thereby highlighting the causal relationships within the model's reasoning. These "what if" scenarios do more than alter outcomes; they enrich the AI's decision-making process, allowing users to explore and understand the model's behavior under various hypothetical conditions. This aspect of counterfactuals in XAI aids in debugging and improving AI systems and also enhances user trust, making AI decisions more relatable and understandable, thereby providing a deeper human-AI relationship (Byrne, 2019).

1.2 Research Problem and Objective

The central challenge addressed by this thesis is the limited explainability in path-based knowledge graph recommender systems. These systems are adept at stating the reasons for selecting a particular product but often lack the capability to delve into hypothetical "what-if" scenarios, such as the impact of different attributes on recommendations. Specifically, the systems do not answer questions like, "What if the recommended product had different attributes?" Understanding which attributes would still make it a plausible choice is not only a highly valuable business insight but also aligns closely with the natural human cognitive process of exploring alternatives.

The main problem in counterfactual analysis lies in the construction of hypothetical scenarios, as the search space of all possible attributes is vast, and exploring all possibilities is computationally expensive, with complexity growing exponentially with the size of the graph. Identifying relevant attributes with a higher likelihood of relevance could transform this otherwise infeasible task into a practical and valuable technique.

To address these limitations, this thesis applies counterfactual analysis to path-based knowledge graph recommender systems, aiming to explore how hypothetical changes to product attributes and interactions could alter the generated recommendations, thereby enhancing the system's explainability. For the challenge of attribute selection, we narrow down the search space by employing community detection techniques and identifying and removing less informative nodes. The primary objectives of this research are twofold:

1. **Development of an Analytical Model:** To create an analytical model that evaluates how changes in the attributes of recommended items affect the recommendation outcomes. This model is designed to identify attributes that would continue to uphold the recommendation's validity, providing deeper insights into the decision-making process of the recommender system.
2. **Marketing and User Experience Insights:** To leverage the insights gained from the analytical model to offer marketing intelligence, foster diversity in recommendations, and formulate user profiles that influence these recommendations. This objective aims to enhance the way businesses understand and cater to diverse consumer preferences, thus improving user satisfaction and engagement.

By meeting these objectives, the research will not only advance the theoretical understanding of explainable AI in recommender systems but also equip businesses with practical tools to refine their marketing strategies and enhance recommendation processes.

1.3 Methodological Approach

The methodological approach developed in this thesis represents a novel strategy for enhancing the explainability of knowledge graph-based recommender systems through counterfactual analysis. This approach employs techniques in data processing and graph analytics to simulate potential scenarios under various attribute conditions. By manipulating attributes and observing the resultant changes in recommendations, this method offers a detailed view into probable user behavior in different circumstances.

Key Components

The methodology consists of several essential components:

1. **Attribute Extraction:** This process involves identifying and extracting relevant attributes and interactions of recommended products from the knowledge graph. This step is crucial for understanding the specific factors that might influence the recommendations.
2. **Community Identification:** Utilizing community detection algorithms helps manage the complexity and size of the data by grouping similar products and attributes. This organization enhances the relevance and efficiency of the counterfactual analysis by ensuring that the simulations are grounded in realistic and comparable scenarios.
3. **Recommendation Scoring:** This component evaluates each hypothetical scenario by calculating the recommendation scores for altered paths. It allows for a comparison with actual recommendation outcomes to determine if a product with changed attributes would still be recommended. This scoring process is integral in providing insights into hypothetical scenarios that could be further used to refine the system.

1.4 Expected Contributions and Outcomes

Contributions to the Field: This research contributes to the fields of recommender systems and explainable AI by introducing a counterfactual framework that incorporates counterfactual reasoning into the recommendation process. This adds a significant layer of depth to the analytical capabilities of existing systems and enhances the explainability of complex recommender systems.

Anticipated Findings: The anticipated results of this research are expected to offer valuable insights from counterfactual analyses, which could shape more effective marketing strategies and product development efforts. This, in turn, should lead to improved customer targeting and enriched product offerings and understanding of the end users behavior.

1.5 Structure of the Thesis

This thesis is organized into several chapters, each serving a specific purpose to comprehensively cover the research undertaken. Here’s a brief outline of each chapter and what it entails:

Chapter 1: Introduction

This opening chapter sets the stage for the research by introducing the topic, stating the research problem, and outlining the objectives. It provides a thorough background on the importance of explainability in knowledge graph-based recommender systems and introduces the concept of counterfactual analysis as a novel approach to address these challenges.

Chapter 2: Literature Review

The literature review chapter explores existing studies and theories relevant to knowledge graph recommender systems and explainable AI. It discusses previous approaches to improve transparency in AI-driven recommendations, evaluates their limitations, and highlights the need for innovative solutions like counterfactual analysis. This review establishes the theoretical foundation for the research and identifies gaps that this thesis aims to fill.

Chapter 3: Methodology

In this chapter, the research methodology used in the thesis is detailed. It describes the development of the counterfactual analysis, including the processes of attribute extraction, community identification, and the computation of recommendation scores. The chapter elaborates on the tools and algorithms employed, the data collection process, and the techniques used to simulate and analyze various counterfactual scenarios.

Chapter 4: Results

The results chapter presents the findings from the application of the counterfactual analysis. It includes detailed discussions on how different attributes and their hypothetical modifications impact the recommendation outcomes. The results are supported by

quantitative data and visual representations to illustrate the effects of attribute changes on the recommendations.

Chapter 5: Conclusion and Future Work

The concluding chapter synthesizes the findings, discussing the implications for both theory and practice. It evaluates the success of the counterfactual analysis in enhancing the explainability of recommender systems and suggests areas for future research. Recommendations for practitioners in the field of recommender systems are also provided, focusing on how they can implement similar methodologies to improve the transparency and effectiveness of their systems.

Chapter 2

Literature Review

2.1 Explainable AI In Knowledge Graph Recommender Systems

Knowledge Graphs (KGs) are effective tools in enhancing the explainability and accuracy of recommender systems. These structured, relational frameworks capture complex interactions among users, items, and their attributes, allowing for more nuanced recommendations coupled with clear, logical explanations. This literature review synthesizes recent advancements in explainable artificial intelligence (XAI) that utilize KGs to illustrate how these technologies can refine recommendation quality, provide layers of transparency to systems and deliver business value in various aspects.

2.1.1 Path-based Modeling

Path-based modeling has emerged as a fundamental innovation in the utilization of KGs for recommender systems. Techniques such as the Knowledge-aware Path Recurrent Network (KPRN) and Path Language Modeling Recommendation (PLM-Rec) illustrate this trend's dynamic nature. KPRN employs LSTM networks to interpret paths of entities and relationships, pointing out those connections that are most influential in understanding user preferences. This method improves the recommendation process by providing a temporal and semantic depth that traditional models lack, allowing for a better prediction of user behavior based on past interactions (X. Wang et al., 2019). On the other hand, PLM-Rec uses a novel approach by integrating natural language processing techniques to

extend KG paths. This model treats paths as sentences, using a language model to dynamically predict and extend these paths within the KG. Such extensions help the system explore new, potentially uncharted areas of the KG, thereby enhancing the system’s ability to recommend items that were previously unreachable. This approach addresses the inherent limitations of static KG structures and improves the system’s recall capabilities, making it particularly valuable for discovering long-tail items (Geng et al., 2022). Together, these path-based methods signify a shift towards more dynamic and exploratory use of KGs, expanding both the depth and breadth of what recommender systems can achieve. Similarly, the Policy-Guided Path Reasoning (PGPR) model uses reinforcement learning to navigate through the KG, selecting paths that not only lead to relevant recommendations but are also interpretable. This model provides explicit paths that detail the reasoning behind each recommendation, fulfilling the dual requirements of accuracy and transparency in the recommendation process (Xian et al., 2019).

2.1.2 Explainability through Symbolic and Logical Reasoning

Monotonic GNNs (MGNNs), introduced by Tena et al. (2022), exemplify the trend of explainability of systems by ensuring that every transformation within the network adheres to a set of logical rules, akin to traditional rule-based systems. This adherence guarantees that the network’s operations are interpretable and justifiable, enhancing user trust by providing comprehensible explanations for the recommendations made.

The integration of user behavioral data into KGs has significantly refined the personalization capabilities of recommender systems. The "Cafe: CoArse-to-FinE neural symbolic reasoning method" model by Xian et al. (2020) represents a sophisticated application of this concept, employing a coarse-to-fine strategy where initially broad user profiles help to narrow down and guide the path-finding algorithms in KGs. These profiles are crafted from historical data and are essential in focusing the recommendation process on paths most relevant to individual users, thus enhancing both the relevance and personalization of the recommendations. This method mirrors strategies used in other models that combine knowledge-base embeddings (KBE) with collaborative filtering. By embedding user behaviors and item characteristics into a unified representation, these models achieve a granular understanding of user-item relationships. This integration allows for a tailored recommendation experience, where the system’s outputs are closely aligned with individual preferences and behaviors, as demonstrated in the work by Ai et al. (2018).

The convergence of these methodologies highlights a crucial trend towards enhancing

both the predictive accuracy and the interpretability of KG-based systems. Through the integration of dynamic path exploration, personalized user profile analysis, and logical reasoning, these approaches offer a more profound understanding of the intricacies involved in making recommendations. They collectively emphasize a shift towards recommender systems that are not only effective in their predictions but also provide transparent and understandable explanations, aligning with the growing user and business demand for transparency and accountability in AI systems.

2.2 Counterfactual Reasoning in XAI

Counterfactuals in Explainable AI provide tool to explore the hypothetical settings. Addressing the challenge of generating sparse and plausible counterfactual explanations, GDPR introduced by Keane and Smyth (2020) introduces a novel case-based reasoning (CBR) approach that uses a curated case-base of effective counterfactuals. In this system, each explanation case serves as a minimal yet impactful alteration of another, resulting in a different class outcome. The method involves identifying and pairing cases within datasets to build an explanation case-base. For new queries, counterfactuals are generated by retrieving and adapting the most similar case from this base, guaranteeing minimal changes and high plausibility. The generated counterfactuals are systematically validated and integrated throughout the experiments to assess the model’s explanatory competence. By using the CBR model, this approach effectively improves the quality and applicability of explanations, meeting the critical need for actionable counterfactuals in AI systems and enhancing their interpretability and trustworthiness.

Further expanding the field of counterfactual explanations in graph-based models, the CF-GNNExplainer model (Lucic et al., 2021) iteratively perturbs the adjacency matrix of a graph, typically through edge deletions, to generate counterfactual scenarios that influence the predictions made by Graph Neural Networks (GNNs). These counterfactuals help evaluate the minimal perturbations needed to effect a change in predictions, employing a loss function that delicately balances the change in prediction with the extent of the perturbation. This nuanced approach facilitates precise modifications to GNN outputs, deepening the understanding of model behavior in graph-based data.

The Causal Knowledge Graph (CausalKG) framework introduced by Jaimini and Sheth (2022) integrates a Causal Bayesian Network (CBN) and a hyper-relational graph

representation using RDF-star to effectively model complex causal relationships. The central neural model, the Causal Bayesian Network, facilitates the manipulation of variables to generate counterfactuals by hypothesizing interventions in the CBN and observing the resultant effects on connected variables. This approach allows the system to predict outcomes under various scenarios. Integrating these counterfactuals into experiments aids in evaluating the causal impacts of different decisions and also ensures that AI outcomes are contextually relevant and more comprehensible, underscoring the value of counterfactual reasoning in making AI systems more transparent and understandable.

The CoCoX model, introduced by Akula et al. (2020), attempts to expand the transparency and understandability of decisions made by deep convolutional neural networks (CNNs). This model generates both conceptual and counterfactual explanations by exploiting cognitive fault-lines—semantically significant features that are critical for classification. CoCoX identifies minimal changes, such as the addition or removal of specific features like stripes or bumps, that could influence the CNN’s output. Based on the VGG-16 architecture, CoCoX utilizes Grad-CAM for effective feature extraction and K-means clustering to pinpoint explainable concepts. These concepts help define the positive and negative fault-lines which are crucial for developing counterfactuals. These counterfactuals are approached as an optimization problem that aims to modify the minimum number of features necessary to change the classification outcome.

For graph-based models, the CF2 framework addresses the unique challenge of enhancing graph-based model explainability by generating counterfactuals that identify necessary and sufficient conditions for predictions made by graph neural networks (GNNs). This framework solves this by integrating factual and counterfactual reasoning into an optimization problem, aiming to pinpoint the minimal sub-graph components whose alteration would change the outcome. Employing a novel loss function and relaxation techniques, CF2 balances the explanation’s complexity and effectiveness, providing iterative refinement to focus on critical graph components that significantly impact the GNN’s predictions. These insights are crucial for advancing the transparency and reliability of GNNs, ultimately enriching the field of explainable AI (Tan et al., 2022). Combined with the aforementioned CoCoX model, these developments underscore the breadth of methodologies being pursued in the realm of explainable AI, reflecting a robust and multi-dimensional approach to understanding and improving the decision-making processes of complex models.

The CLEAR framework innovatively generates counterfactual explanations for graph-level prediction models by utilizing a graph variational autoencoder (VAE). This approach

involves encoding the node features and graph structure into a latent space, and then decoding from this space to construct counterfactual graphs that minimally alter the original while achieving a specific prediction outcome. By mapping graphs into a latent space, CLEAR allows for both effective optimization and generalization across graphs, while an auxiliary variable enhances the model’s ability to adhere to underlying causal relationships. This method advances the generation and application of counterfactual explanations in graph data, outperforming existing techniques in validity, proximity, and causality (Ma et al., 2022).

Z. Wang et al. (2021) address the challenge of enhancing knowledge graph reasoning by employing counterfactual scenarios to identify crucial relationships within the graph. To tackle this issue, the approach involves generating counterfactuals by modifying relationships in factual reasoning paths and observing the resultant impacts on reasoning outcomes. These modifications help determine the importance of each relationship, which are then assigned weights. This weighted information is integrated as prior knowledge into a reinforcement learning-based reasoning model. Specifically, it utilizes a policy network that combines these weights with neural outputs to guide decision-making. The integration of counterfactual-derived weights with the neural model significantly enhances the reasoning capabilities of the system, as demonstrated through robust experimental validation across multiple large datasets. This method boosts performance and aids in maintaining consistent reasoning across varied path lengths, thereby enriching the explainability and reliability of knowledge graph-based systems.

Incorporating counterfactual reasoning with knowledge graph completion (KGC), the task of CounterFactual Knowledge Graph Reasoning (CFKGR) explores hypothetical alterations within a knowledge graph. The study introduces a neural model called COULDD (COUnterfactual Reasoning with KnowLedge Graph EmbeDDings), which adeptly refines existing knowledge graph embeddings to effectively manage hypothetical scenarios. Through the extraction of logical rules from the knowledge graph, counterfactuals are generated, creating scenarios that involve adding new edges and potentially removing contradictory ones. These scenarios are then assimilated by updating the embeddings with counterfactual information and re-training the model to discern the validity of these new configurations. This innovative methodology enables the model to evaluate the legitimacy of counterfactual changes and boosts its ability to navigate potential scenarios, thereby maintaining precision (Zellinger et al., 2024)

2.3 Counterfactual Methods in Different Types of Recommender Systems

Counterfactual reasoning in recommender systems has emerged as a technique within the domain of explainable artificial intelligence (XAI), enhancing both the transparency and fairness of recommendations. By modeling alternative scenarios where specific variables are modified, this approach provides insights into the potential impacts of different data configurations in recommendation process.

The introduction of the KGCR model (Y. Wei et al., 2023) marks an advancement in embedding causal inference within graph-based recommender systems. Using Graph Convolutional Networks, this model improves user, item, and attribute embeddings, which allow for a more nuanced understanding of user preferences. By constructing a causal graph and applying do-calculus interventions, the KGCR model effectively mitigates biases introduced by previous user interactions, offering a refined approach to understanding how bias influences recommendation outcomes.

In a similar vein, Tran et al. (2021) developed the ACCENT framework, which facilitates the generation of actionable counterfactual explanations in neural recommender systems. This framework leverages extended influence functions to explore how changes in user-item interactions could affect recommendation outputs, enhancing computational efficiency through Fast Influence Analysis. This methodology underscores the minimal adjustments in user behavior that could lead to different recommendations, thereby aiding in the creation of more transparent recommendation mechanisms.

Addressing selection bias, (Liu et al., 2022) implemented counterfactual policy learning to recalibrate recommendation fairness and effectiveness. Their approach uses Inverse Propensity Scoring to weigh observed interactions, allowing the system to simulate outcomes under different recommendation policies. By integrating these counterfactual outcomes into the learning process, the model achieves an improved balance, enhancing both the performance and equity of recommendations across various user groups and item categories.

The Prince method (Ghazimatin et al., 2020), emphasizes the importance of trust and understanding in recommendation systems through counterfactual reasoning within heterogeneous information networks. By identifying key user actions and employing Personalized PageRank, Prince efficiently predicts the impact of these actions on recommendation outcomes. This approach avoids exhaustive computations outperforms traditional

heuristic methods in providing understandable and trust-enhancing explanations.

Finally, the Counterfactual Explainable Recommendation (CountER) model (Tan et al., 2021) focuses on identifying minimal attribute changes that could reverse a recommendation decision. Through a structured optimization process, CountER iteratively adjusts item attributes to discover the least extensive yet impactful changes required for altering outcomes. This model uses novel metrics to evaluate the necessity and sufficiency of these changes, demonstrating enhanced precision in providing actionable insights into recommendation decisions.

While counterfactual reasoning enhances the transparency and fairness of recommender systems, it also introduces potential vulnerabilities that can be exploited. A notable exploration of this issue is presented in the paper by Chen et al. (2023), titled "The Dark Side of Explanations: Poisoning Recommender Systems with Counterfactual Examples." This study reveals how counterfactual explanations (CFs) can be manipulated to deceive recommender systems. By employing a novel poisoning technique named H-CARS (Horn-Clause Attacks to Recommender Systems), which utilizes a neural model called Neural Collaborative Reasoning (NCR), the paper illustrates a method where logical reasoning through Horn clauses simulates the decision-making processes of recommender systems. Here, CFs are generated by pinpointing the minimal adjustments necessary in user-item interactions to alter the outcome of recommendations. These counterfactuals are critical in training a surrogate model, which then crafts targeted item embeddings and simulates user interactions to deceive the system. This approach highlights the application and generation of CFs and also emphasizes their potential for both analysis and exploitation, particularly within environments sensitive to security.

The Counterfactual Explainable Recommendation (CERec) system by X. Wang et al. (2024) employs reinforcement learning to navigate and optimize paths in a knowledge graph, focusing on attribute-based counterfactual explanations. It identifies counterfactual paths where minimal attribute adjustments lead to notable shifts in recommendation outcomes. An essential feature of CERec is its adaptive path sampler, which incorporates a two-step attention mechanism to efficiently manage the extensive search space of the knowledge graph. This selective exploration ensures computational efficiency and enhances the relevance of the explanations provided. Counterfactual scenarios are crafted by simulating changes in item attributes and observing the consequent alterations in recommendation outputs. These insights are then reintegrated into the recommendation model, improving its accuracy and reducing the incidence of undesirable recommendations. This

makes CERec a useful tool in enhancing both transparency and decision-making in recommendation systems, contributing to more accurate and user-satisfying outcomes.

In conclusion, counterfactual reasoning offers a robust framework for enhancing the explainability and fairness of recommender systems by providing a deeper understanding of the implications of various data interactions and policies. These innovative approaches clarify the decision-making processes and provide more equitable and user-centric recommendation practices.

2.4 Application of Counterfactuals for Fairness and Bias Mitigation

One of the uses of counterfactual reasoning is to mitigate biases in automated decision-making systems. This method involves hypothesizing alternative scenarios where key variables are altered, exploring how such changes impact outcomes. By doing so, it uncovers hidden biases and ensures fairness in AI operations. Widely applied across various AI frameworks, from graph neural networks to recommender systems, counterfactual reasoning is considered an essential tool for ethical AI development.

The use of counterfactual reasoning in graph-based models like those studied Z. Guo et al. (2023) demonstrates a rigorous approach to maintaining consistency in model predictions across varying sensitive attributes, which is necessary to guarantee fairness. By implementing Graph Variational Autoencoders (GraphVAE), they not only perturb attributes but also train the network to minimize discrepancies in outputs between the original and counterfactual nodes. This methodology effectively addresses biases at a fundamental level, ensuring the fairness of the model's outcomes. Medda et al. (2024) extend this approach within graph neural network-based recommender systems. Their innovative use of counterfactual reasoning to adjust user-item interactions on a bipartite graph includes strategically adding or removing connections, which serves to simulate various scenarios where demographic disparities can be analyzed and mitigated, ensuring a more equitable distribution of utility among users. The field of recommender systems frequently grapples with biases such as popularity and exposure, which can distort user preferences. T. Wei et al. (2021) dissect these issues through the Model-Agnostic Counterfactual Reasoning (MACR) framework, which explicitly separates the influence of item popularity from actual user preferences. By adjusting input data to simulate a scenario

where item popularity is neutralized, MACR provides a recalibrated basis for recommendation, aligning more closely with unbiased user preferences. Meanwhile, Xu et al. (2020) focus on exposure bias by employing a counterfactual approach that involves a minimax adversarial model. This model simulates worst-case scenarios to test the resilience of the recommendation system, ensuring that it can withstand and adapt to a range of user exposure conditions, thus promoting a more fair and balanced recommendation landscape.

Addressing data sparsity and imbalance, Yang et al. (2021) utilize causal inference via Structural Equation Models (SEMs) to generate counterfactual scenarios that enrich training datasets. This addresses the immediate issue of insufficient data and also simulates a broader spectrum of user interactions, which helps in developing a more robust and responsive recommender system. On a more focused level, Chiappa (2019) experiments with the use of Path-Specific Counterfactual Fairness (PSCF) within decision-making processes. This approach manipulates causal pathways, particularly those that might be influenced by sensitive attributes such as race or gender, to ensure that resulting decisions are free from the undue influence of these attributes, thus promoting fairness in critical decision-making contexts.

2.4.1 Leveraging Knowledge Graphs for Fair Recommendations

Expanding the utility of knowledge graphs, Balloccu et al. (2022) integrate counterfactual reasoning within the Policy-Guided Path Reasoning (PGPR) model to optimize recommendation systems. By re-ranking items and explanations based on various fairness-oriented criteria, such as recency, popularity, and diversity, PGPR enhances the quality and equity of recommendations. This approach not only improves the relevance of the recommendations but also significantly increases user trust and satisfaction by ensuring that recommendations cater equitably to diverse user groups.

Chapter 3

Research Design and Methodology

This chapter details the methodology employed to achieve the primary objective of this study: enhancing the explainability of path-based knowledge graph-based recommender systems by narrowing the search space through community detection and outlier node recognition, as a result enabling the exploration of "what-if" scenarios within a feasible scope.

The methodology is structured as follows:

1. **Recommendation Engine/System Utilization:** For the recommendation engine, we employ CAFE (Coarse-to-Fine Neural Symbolic Reasoning for Explainable Recommendation). This system is particularly suitable for our purposes due to its path-based nature and its capability to evaluate the plausibility of different paths by assigning probability scores to the steps that connect users to products.
2. **Counterfactual Analysis:**
 - **Initial Recommendation:** The process initiates with the system generating a product recommendation for a user. In a path-based recommendation system, the built-in explanation typically includes a sequence of entities and relationships that link the user to the recommended product. Then, the user selects which product recommendation they want to analyze with counterfactual analysis.
 - **Extraction of Relevant Information:** Relevant attributes for the counterfactual analysis are those that could potentially be associated with the entity under examination, which have a higher likelihood of achieving a score above the threshold and being verified as plausible compared to the rest of the attribute search space in the knowledge graph. The process of extracting relevant

information begins with the removal of outlier nodes in the knowledge graph, using centrality measures to identify these nodes. Based on the nodes along the path, the recommended product, and connected entities, we select potential relevant attributes. These attributes are then further refined by considering the communities identified within the graph.

- **Scenario Construction:** Utilizing the extracted information, a collection of hypothetical scenarios are constructed. These scenarios are crafted to test various alterations in attributes and their impact on the recommendation outcome.
- **Verifying The Plausibility of Scenarios**

3.1 Recommendation Engine/System Utilization

The approach suggested in this thesis is designed for any knowledge graph-based recommender system that is path-based and scores the path at each step. We have found the CAFE (Coarse-to-Fine Neural Symbolic Reasoning for Explainable Recommendation) methodology to be a suitable foundation to work with for our objectives.

CAFE forms the basis of our path-based knowledge graph recommender system. This methodology is particularly suited to our needs because it allows for detailed access to the scores of each path within the graph, facilitating the analysis of counterfactual scenarios. This section provides an overview of its implementation and core functionalities.

3.1.1 Data and Implementation

The CAFE model is implemented using the Amazon review dataset, the beauty category, which includes comprehensive user and product interactions. It leverages predefined embeddings train in the model developed by Ai et al. (2018) described in the previous section, as input to their symbolic network.

3.1.2 Knowledge Graph Composition

The knowledge graph at the heart of this recommendation system is intricately structured, comprising several types of entities and relationships:

- **Users** are linked to the products they have purchased and the words they have used.

- **Products** are associated with descriptive words, their brand, category, and other related products. Relationships with related products include those that have been 'bought together', 'also viewed', and 'also bought'.
- **Brands and Categories** are the specifications of the products.
- **Related Products** mentioned above.

The entity types and the relationships between the nodes are demonstrated in figure 3.1.

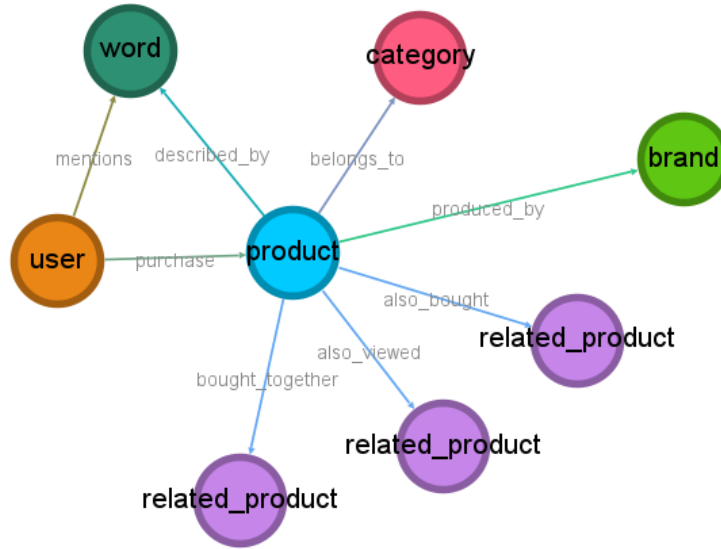


Figure 3.1: Knowledge Graph Nodes and Relations Types

3.1.3 Path-Based Recommendation Mechanism

The system operates on predefined metapaths that represent meaningful relationships leading a user to a product. These metapaths outline the possible structure of paths that could lead a user to purchase a product. The structures of metapaths used in the recommender system at hand are demonstrated in figure 3.2 on the next page.

The recommender system assigns a probability score to each step along the path, determining the strength of the connection between the user and the potential product recommendations. It then selects the top 10 paths with the highest cumulative scores, and the products associated with these paths are recommended to the user. This scoring and selection process ensures that the recommendations are both relevant and tailored to the user's preferences and behavior patterns. This structured approach allows the CAFE

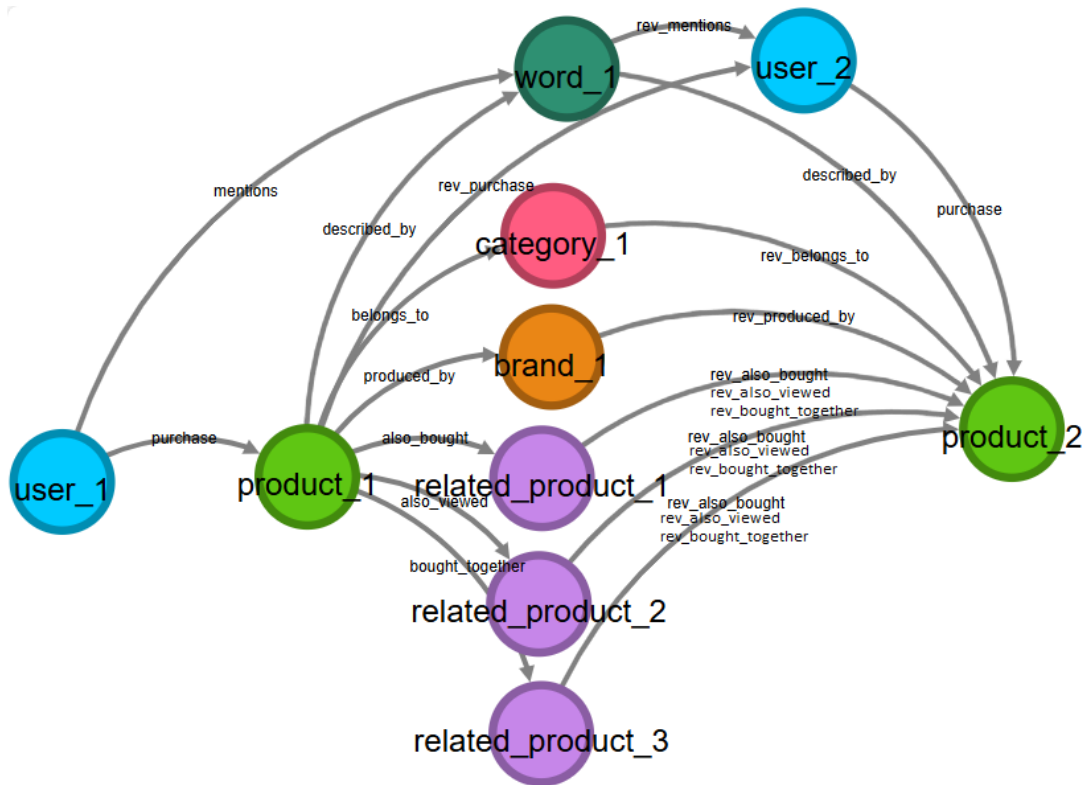


Figure 3.2: Metapaths Defined Within Recommender System

system to effectively recommend products and offer insights into the reasons behind each recommendation, enhancing the system's explainability.

3.2 Counterfactual Analysis

Initial Recommendation

3.3 Extraction of Relevant Information

To select the most relevant counterfactual attributes for the product under analysis and ensure they add value to our study, we eliminate outlier nodes from the knowledge graph. Additionally, we conduct community detection to restrict attribute selection to those within the same community as the product being analyzed.

Outlier Nodes Detection and Removal

For the purpose of the outlier detection and removal, we use the degree centrality of nodes in the knowledge graph. Nodes that exhibit a degree centrality above a certain threshold are eliminated. While these highly connected nodes can provide valuable insights for counterfactual analysis, they often include unhelpful elements such as universally common categories or frequently used words that lack distinctive features. Additionally, since they disrupt the modularity of communities essential for selecting relevant attributes, these nodes are removed.

Degree Centrality of Nodes

For a given node, degree centrality is the number of edges connected to that node (**bloch2023centrality**)

The use of degree centrality for outlier detection is justified given the structure of the knowledge graph we have utilized, where node types such as category, brand, and related products are connected exclusively to product nodes, as shown in Figure 3.1. This measure will determine how many products each of these attribute nodes is connected to, which will help in deciding whether to retain or remove them as outliers based on a predetermined threshold. For word attributes, however, since word nodes are connected not only to product nodes but also to user nodes, we apply a modified approach. Rather than using degree centrality, we count only the connections between each word node and product nodes.

The degree centrality of different types of nodes are demonstrated in figure 3.3 on the facing page. It is apparent from the plots that some nodes have unusual degree centralities.

Based on the reasons described, it is necessary to remove the outlier nodes while retaining the structure of the original knowledge graph as much as possible. Various statistical methods for outlier detection were experimented with, and the method that removed the least number of nodes while maintaining the highest community resemblance is chosen, as preserving the original data and structure as much as possible is preferred. Specifically, the knee method is used to identify and remove the outlier nodes.

Knee Detection

For the detection of the knee in the degree centrality of nodes, the Kneedle algorithm have been used. The Kneedle algorithm identifies "knee points" in data, which represent

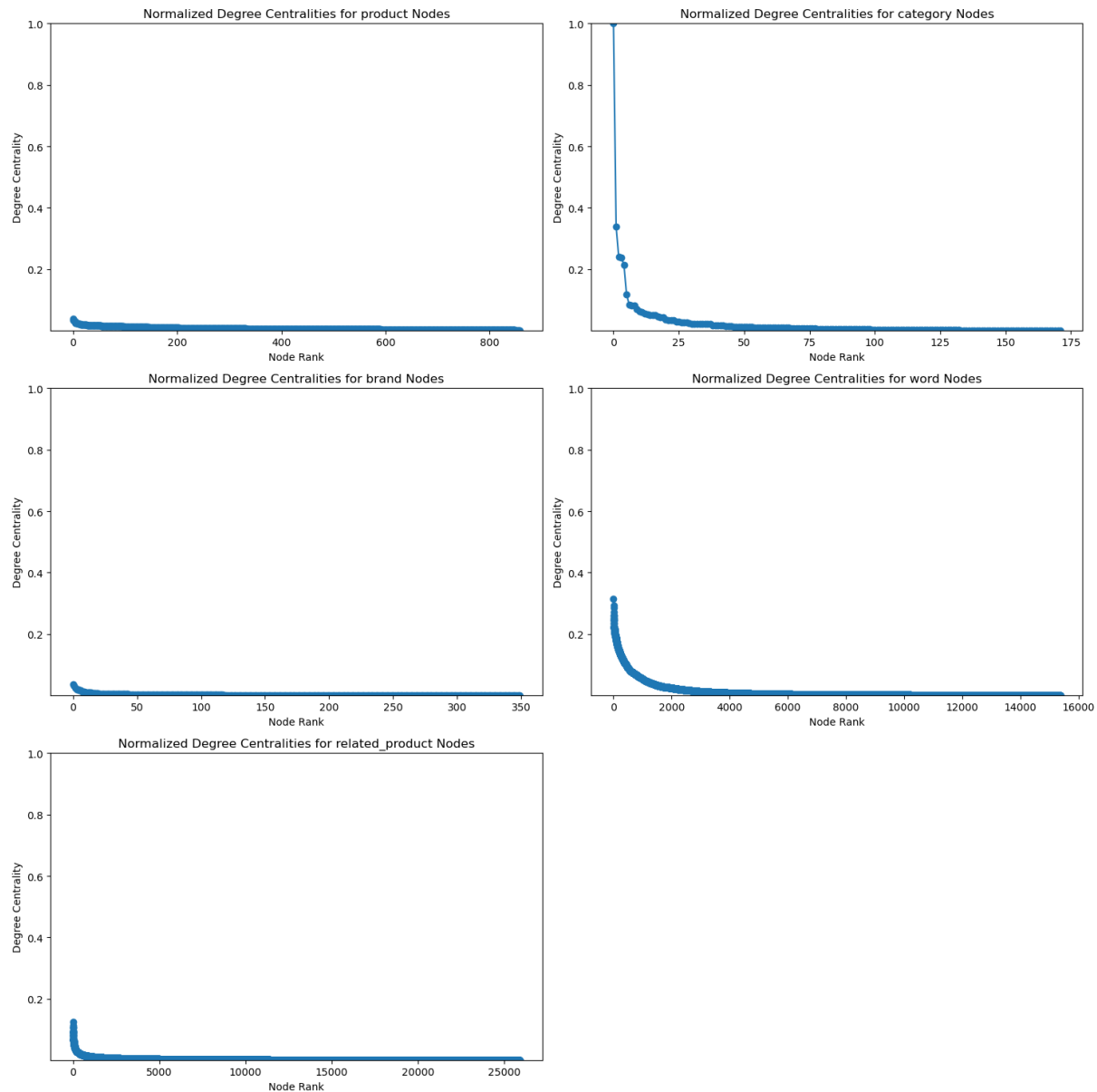


Figure 3.3: Degree Centrality Distributions of Nodes by Node Type

the points where the rate of improvement in performance begins to level off significantly in relation to changes in a tunable system parameter, degree centrality in our case. It operates by first calculating a difference curve from normalized data points, highlighting the deviations from a direct diagonal line. The algorithm then detects local maxima in this difference curve, pinpointing where the rate of change in the response variable starts to decrease sharply, indicating the presence of a knee ([satopaa_finding_2011](#)).

In figure 3.4 the degree centralities of all nodes with different types are represented in a plot, arranged in decreasing order. The Kneedle algorithm is used to detect the knee point in the values of degree centralities. This knee point marks a transition where the differences in degree centrality become less significant beyond this point compared to those before it.

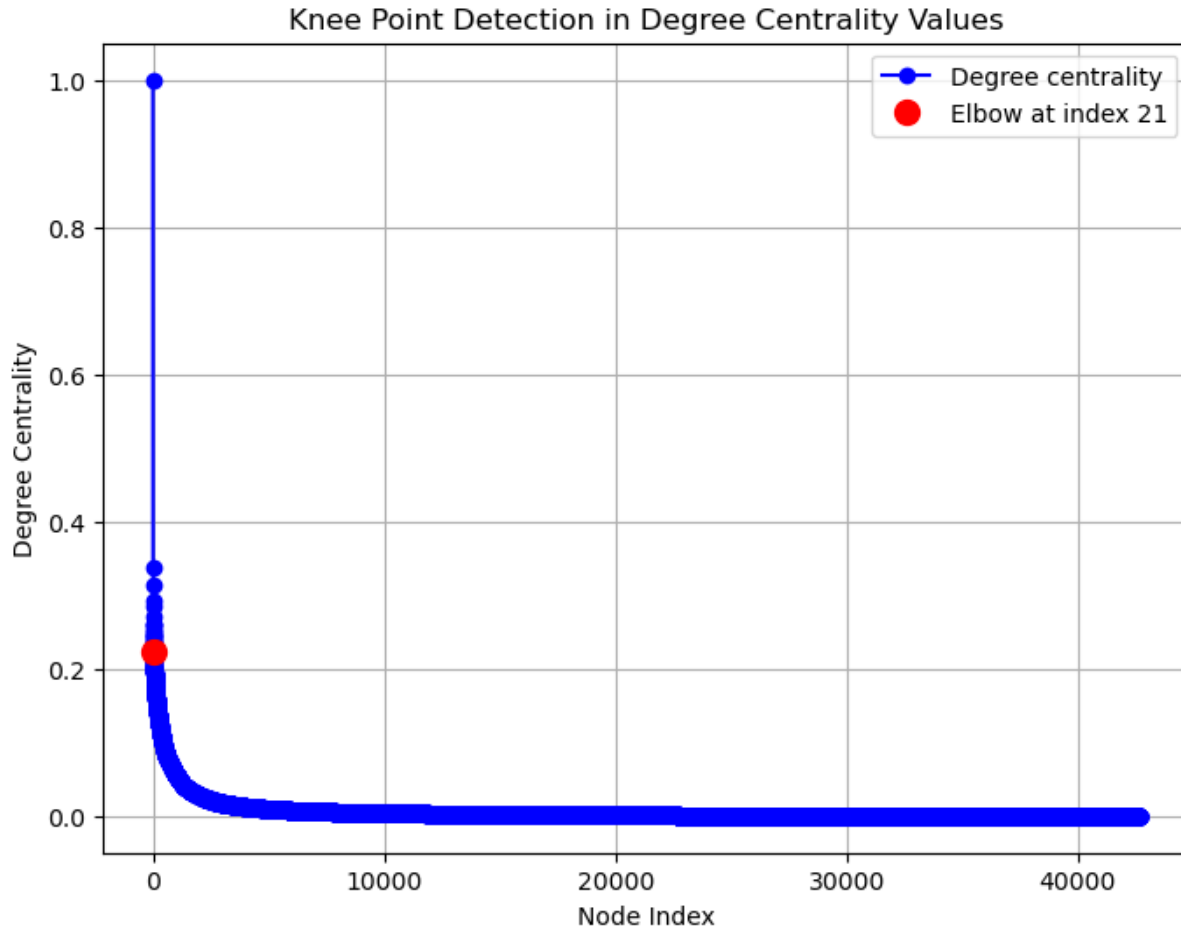


Figure 3.4: Degree Centrality of All Nodes and Knee Point Specified with Kneedle Algorithm

In this study, the Kneedle algorithm is applied to the degree centrality of all node types collectively, rather than separately, to ensure a consistent and meaningful outlier removal process. Since the goal is to eliminate nodes with excessively high degree centrality while retaining the overall structure of the knowledge graph, a global knee point is more appropriate. Applying knee detection separately for each type could lead to over-filtering, especially for node types with lower variability, potentially unnecessarily removing nodes

that are still structurally important. By considering all node types together, the method ensures that only the most extreme outliers are removed while preserving the connectivity and integrity of the network.

In table 3.1 the number of nodes removed and consequent changes are represented.

Metric	Value
Nodes Removed	25
Number of Communities in Original Graph	22
Number of Communities in Cleaned Graph	24
Number of Nodes in the Original Graph	42670
Number of Nodes in the Cleaned Graph	42645

Table 3.1: Graph Metrics Before and After Cleaning

Community Detection and Graph Analysis Based Node Filtering

The next step in the counterfactual analysis is the detection of communities within the knowledge graph of interactions. Communities are identified using Louvain Method. This helps to cluster entities that share significant similarities and interactions. The Louvain Method is a an efficient algorithm designed for detecting communities in large-scale networks by optimizing modularity, a measure that quantifies the density of links within communities relative to those between them introduced in (**blondel_fast_2008**).

In the case of weighted networks the modularity Q is defined as:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \quad (3.1)$$

where A_{ij} represents the weight of the edge between nodes i and j , $k_i = \sum_j A_{ij}$ is the sum of the weights of the edges attached to vertex i , and c_i is the community to which vertex i is assigned. The δ -function $\delta(c_i, c_j)$ equals 1 if $c_i = c_j$ and 0 otherwise. Here, $m = \frac{1}{2} \sum_{i,j} A_{ij}$ represents the sum of the weights of all the edges in the network. In the case of unweighted networks, where no specific weights are assigned to the edges, each edge is assumed to have a weight of 1.

The change in modularity ΔQ when moving a node i to a different community is given by:

$$\Delta Q = \left[\frac{\sum_{\text{in}} + k_{i,\text{in}}}{2m} - \left(\frac{\sum_{\text{tot}} + k_i}{2m} \right)^2 \right] - \left[\frac{\sum_{\text{in}}}{2m} - \left(\frac{\sum_{\text{tot}}}{2m} \right)^2 - \left(\frac{k_i}{2m} \right)^2 \right], \quad (3.2)$$

where $\sum_{\text{in}}$ is the sum of the weights of the edges inside the community, $k_{i,\text{in}}$ is the sum of the weights of the edges from node i to other nodes in the community, and $\sum_{\text{tot}}$ is the total sum of the weights of the edges to nodes in the community. k_i represents the degree of node i , which is the sum of the weights of all edges attached to node i . If the network is unweighted, the weight A_{ij} is taken to be 1 for all edges.

The algorithm operates in two iterative phases: initially, it optimizes modularity locally, evaluating potential gains by moving individual nodes into different communities. Nodes are shifted to the community that maximizes this gain, and the process is repeated until no further improvement is possible, achieving a local maximum of modularity. In the second phase, the method aggregates these identified communities into new nodes of a reduced network, and the process is reapplied. This hierarchical approach allows the algorithm to uncover community structures at multiple levels effectively. Notable for its speed, the Louvain Method can handle networks with up to millions of nodes efficiently, making it well-suited for modern datasets of substantial size. (**Blondel_2008**)

Verifying The Plausibility of Scenarios

For the counterfactual analysis of the predictions provided by the recommender system, first a threshold is determined for the minimum score required for a product path to be recommended to a user. This threshold corresponds to the path score of the last product in the top 10 recommended products. This method of setting the threshold is based on the idea that in a recommender system suggesting 10 items, any item ranked below the 10th is not recommended. Therefore, the score of the 10th item serves as a natural cutoff, as any lower score would fall outside the recommendation list. This approach is commonly used for defining thresholds in counterfactual analysis in recommender systems.

The next step involves retrieving the first-level products related to the attributes of the recommended product, and through them, extraction of possible relevant attributes. This forms the initial layer of attribute selection, grounded on the hypothesis that first-level connected items possess possibly relevant attributes. These attributes are then filtered based on the community of the recommended product, meaning the attributes that are present in the same community as the product recommended, were used for the Construction of counterfactual scenarios. This ensure more relevance of the attributes used in the counterfactual scenarios.

Counterfactual Scenarios

Since the scoring of attributes is based on the metapaths defined in the system, after the selection and filtering of relevant attributes, an appropriate metapath is selected based on the type of the attribute. Using the recommender engine, we calculate the score for user-product combinations, which incorporates all the products previously purchased by the user and the counterfactual attribute in question. If the recalculated score for a product, involving the counterfactual attribute, exceeds the determined minimum score, the attribute is considered a positive influence for a product similar to the recommended one.

This isolated attribute analysis aids in understanding the influence of specific attributes on product recommendations and also provides marketing insights. Such insights can further be used to enhance the diversity and precision of the recommender system.

Chapter 4

Results

This chapter presents the results from applying the counterfactual analysis methodology to a path-based knowledge graph recommender system. We start by examining the initial recommendations generated by the recommender system for a random user. Following this, we explore the outcomes of our counterfactual analysis to understand how different attributes affect product recommendations.

4.1 Case Study

To provide a concrete example of our counterfactual analysis methodology process, we will conduct a case study. This case study will illustrate the application of our methodology, demonstrating its effectiveness and potential insights.

4.1.1 Initial Recommendations

To initiate our analysis, we first examined the top 10 recommendations generated by the CAFE recommender system engine for a specific user (user 0). These recommendations, along with their associated paths and log probabilities, provide insight into the system's decision-making process. Below, we present these recommendations and discuss their implications.

Top 10 Recommendations for User 0:

1. `([0, 261, 849],
[-4.5635014, -3.4569123],
["mentions", "rev_described_by"])`

2. ([0, 9826, 123],
[-5.4080114, -3.6974714],
["mentions", "rev_described_by"])
3. ([0, 14974, 396],
[-5.5097566, -3.6836069],
["mentions", "rev_described_by"])
4. ([0, 6797, 528],
[-5.648406, -3.7231576],
["mentions", "rev_described_by"])
5. ([0, 856, 455, 330],
[-2.7392018, -1.455907, -3.5228138],
["purchase", "also_viewed", "rev_also_bought"])
6. ([0, 856, 479, 626],
[-2.7392018, -2.2539396, -3.566442],
["purchase", "also_viewed", "rev_also_bought"])
7. ([0, 856, 4350, 624],
[-2.7392018, -8.304826, -4.138239],
["purchase", "described_by", "rev_described_by"])
8. ([0, 856, 13745, 241],
[-2.7392018, -8.540054, -4.213147],
["purchase", "described_by", "rev_described_by"])
9. ([0, 856, 3327, 793],
[-2.7392018, -8.5651655, -4.2710376],
["purchase", "described_by", "rev_described_by"])
10. ([0, 856, 3, 45],
[-2.7392018, -0.89929277, -3.0066423],
["purchase", "produced_by", "rev_produced_by"])

Analysis of Recommendations:

Path Types: We observe four distinct path types in these recommendations:

- 'mentions' → 'rev_described_by' (4 instances)

- 'purchase' → 'also_viewed' → 'rev_also_bought' (2 instances)
- 'purchase' → 'described_by' → 'rev_described_by' (3 instances)
- 'purchase' → 'produced_by' → 'rev_produced_by' (1 instance)

The probability scores (represented as negative log-likelihoods) [placeholder for logo probs definition] vary significantly across the recommendations. Lower absolute values indicate higher probabilities.

In the following sections, we will explore how modifications to these paths and attributes might alter the recommendation outcomes through our counterfactual analysis.

4.1.2 Counterfactual Analysis for User 0's Recommendations

Below is an example of a recommended path that user 0 is recommended product 45 based because of the brand of their previous purchases:

```
([0, 856, 3, 45],
 [-2.7392018, -0.89929277, -3.0066423],
 ["purchase", "produced_by", "rev_produced_by"])
```

That with IDs converted to their corresponding real name for the attribute, it represents:

```
user_0 → purchases → product_856 → produced_by → brand:Maybelline
      → has_produced → product_45
```

After removing the outlier nodes and determining the communities, as illustrated in figure 4.1 on the next page, we select the first-layer attributes that are connected to a product that, in turn, is connected through the attributes of the recommended product. This selection is made for all entities connected to the recommended product. Based on these attributes, counterfactual scenarios are constructed as described in the previous section, and the recommender engine evaluates their plausability.

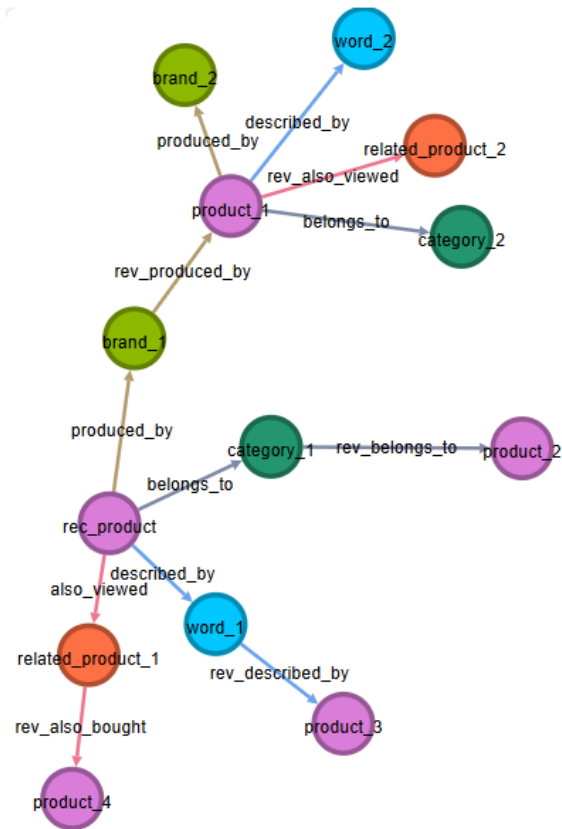


Figure 4.1: Demonstration of First Layer Attribute Selection

We consider the score of the path leading to the 10th recommended product as the minimum acceptable threshold. If a counterfactual path achieves this score, it indicates a scenario where the recommender engine assigns a score suggesting that the user is likely to appreciate the recommendation.

figure 4.2

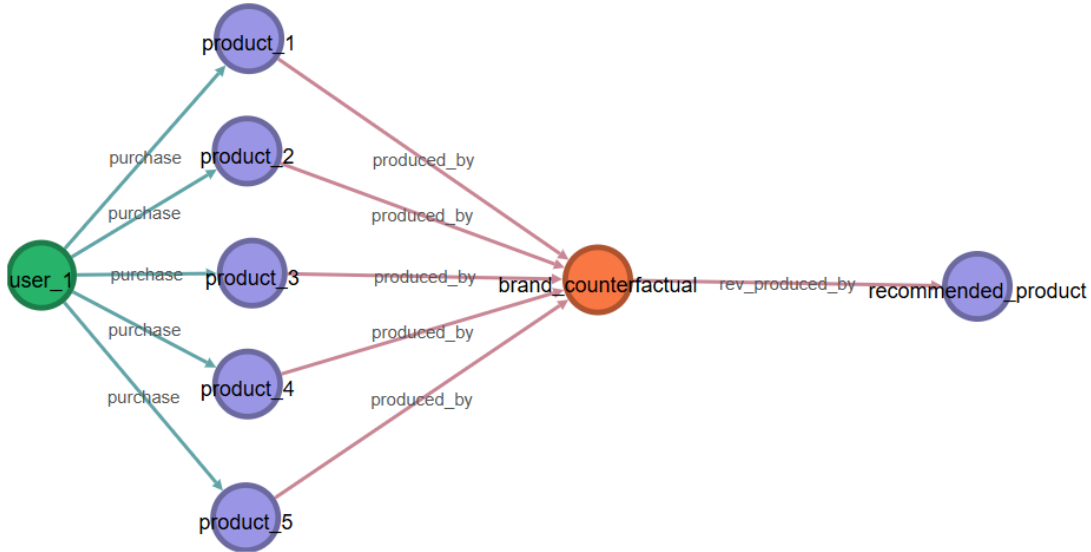


Figure 4.2: An Example of the Construction of Counterfactual Paths for an Attribute Evaluation

The original attributes of the product in this recommendation are:

Words the product is described by:

marker	lipstick	touch	fabulous
anybody	liquid	satisfy	full
iti	wink	shine	precisely
bee	colorsensational	mine	morning
gosh	usage	stiff	point
drawing	downwards	sun	mail
record	kinda	productmy	beyond
gloss	eh	lipcolor	yes
greatit	z	hubby	ordered
rim	berry	regular	remind
lightit	part	turn	immediately

colorstay	wonder	bare	shipping
pop	fricken	chemical	running
ahead	somewhat	dissapointe	fix
sparkle	tint	tacky	delivery
pleasedi	delicious	reapp	intended
rosy	colour	sigh	lasting
nude	outside	itll	bag
folks	ago	ugly	moist
edge	sorta	eos	fade
insult	onto	periodical	amaze
coloring	rather	cost	pieces
itit	medicate	fact	ready
experienced	fuscia	luv	hurt
ohh	matte	toilet	recieve
lipline	taste	mouth	productit
cherry	maybelline	application	properly
stung	assume	complaint	okay
senior	effect	waist	bright
cracked	loving	send	vibrant
indicative	nature	luck	raspberry
wine	palest	exactly	sting
stain	cooltone	else	uncomfortable
lid	watery	kiss	pen
arrive	slightly	took	eat
person	tempt	bother	koolaid
tube	hooked	placed	matter
information	mom	close	overit
name	pigment	must	broken
dayin	medium	alot	orange
buff	gave	m	sell
trash	total	wish	scrub
least	dobut	appear	means
lipstickthe	layer	stroke	certainly
weird	place	peel	obviously
waste	goodnot	drink	cake

putt	understate	practical	liner
picture	chapstick	inevitable	guess
eaten	low	making	sharpie
reorder	future	true	cap
tip	gorgeous	defiant	tend
olay	balm	sum	darkish
judge	damage	n	working
manner	mention	reason	open
lipstain	applied	saturday	thumb
doit	seller	seal	wild
star	inflexible	comfortable	napkin
convenient	clump	poor	paper
mys	bottleit	funky	crayola
inexpensive	roll	fall	barely
felttip	pick	faith	fruit
unfortunately	yo	pinkre	
soon	magic	worst	

Brand of the product is:

Maybelline

Categories the product belongs to:

Makeup	Lips	Lip Stains	Beauty
--------	------	------------	--------

Related Products that have been bought together, also bought, or also viewed with this product:

B006UEUJII	B00HQLC91E	B00303STIY	B00I4DRRXI
B004O3YCQ2	B0045KLPI2	B002YG5IZA	B00BEWRDUI
B00HF6S88I	B0051TSKTS	B00CTONUL6	B00HWT32SE
B004Y9LAW0	B004EFHY0Q	B002YGB8D6	B002YGBNSG
B00HUAOE2S	B006UEUJV0	B00HF7I6ZM	B00H3DAWO6
B00EYZR3E8	B00HWSNF00	B006HCJXBM	B00A86SDLO
B00383O43U	B006UEUHXX	B0082D9R1K	B006UET4XE
B00IR6HW5A	B00F9AALTG	B00HWSNAPU	B002YGDCNU

B004ECH328	B004BCXAM8	B005CYK9HS	B00F0KWBCU
B004KOSZTK	B00CTOLDEC	B007GMJ8Z8	B006XLA6U4
B006UEUIJS	B00HUAPZK8	B006UEUHLC	B009HVMVLY
B006UEUIUC	B004EF6DQC	B0051TSL10	B00FZF3C46
B00BSIPWS8	B009FAPVJG	B004EF8IGU	B002YG6048
B006K5J3WA	B004Y9LNEK	B00H2MSAVK	B00JULSPA2
B006UEUK82	B004679FRM	B00HDE0Q00	B002YGF5FI
B00AM3Z2XU	B004O3YCCL	B005CYK6MG	B00383Q11I
B00383KCCW	B001TK1I4M	B004O3ZPMW	B002YGF5I0
B008GOR6O0	B004EFJW06	B00HWT1A8S	B005OGSO3K
B00FORNVGO	B0045KUO8E	B008B2Z94Q	B004ECGPMM
B008JAPA7Q	B002YGB9JY	B004Y9JVWG	B006XM9M58
B006UEUGWM	B004ECC24W	B004Y9LMN2	B005CYJKAU
B00F0KHUDD	B004CKWWOQ	B0082D9RYM	B004ECIW3C
B005CYKJHI	B00HUAO7ZM	B006UETDR6	B0030GTYM6
B0051TSLOW	B006UEUI8E	B004Y9LM8C	
B006UEUH80	B00HDE1672	B00HU8W2BK	
B0082D9SUK	B0057WJ1N8	B002YGF4QS	

And the plausible attributes approved by the recommender engine are as follows:
BRAND: If the product had one of these as **BRAND**, it would still be recommended through that brand:

HEY	loel	Physicians Formula	Neutrogena
leegoal	Kevyn Aucoin	Epower Mall	Sigma Beauty
e.l.f. Cosmetics	SHANY Cosmetics	Starry	Crazy Cart
Coastal Scents	Max Factor	COVERGIRL	Ailunce
Mehron	NYX	L'Oreal Paris	Maybelline
Jordana	Godefroy	Lime Crime	divaderme
Bonamart	Sleek	Sparkling Beauty	Cover Girl
MISSHA	Essie	Benefit Cosmetics	Wonder Wedges
Hard Candy	Kingmys	Revlon	

Related Products: If any of these **Related Products** were connected to the recommended product, the product would still be recommended through that related product:

B001H270RQ	B005BF1M10	B004BCX8B6	B0013L3XMM
B004Y9GTYE	B002ULRAQE	B001KYNZM0	B00FAEOCP0
B00B1ZRTP2	B004Y9M81W	B00303STIY	B001IB0NOS
B00EL1HZ34	B006UET4XE	B00112BUBO	B006UETDGC
B00BMW24TU	B000CECSAO	B00150LT40	B00KVBZAXK
B002UL5ZKC	B008FX7B9M	B004B3YC9M	B008GOR6O0
B00HF6SJYV	B000US0804	B005OZH88I	B004Y9GV60
B00B1DRA94	B001KYQA1S	B002UL10M4	B00FORNVGO
B008WG7X8Q	B006UEUI8E	B009ZB6MKM	B005UKZAJG
B003K16T06	B0030GTYM6	B006OZEYR0	B003FKSVMG
B002ZTT9II	B002ISV7B8	B004BCXAM8	B009WI3RZQ
B00HWSNF00	B001HKR6WM	B004Y9LM8C	B005EIIEQA
B002WTC38O	B004Y9LAW0	B0048KSGZO	B006OZF3DY
B009K7X26I	B004X1QY20	B00DHQCJTO	B004Y9GVZG
B009HVMVLY	B0013TM9UQ	B004M7WKKK	

WORDS: No word was verified, as if it was connected to the recommended product, the product would be recommended through that word.

The plausible scenarios work in favor of the explainability by demonstrating how changes in specific product attributes, such as the brand or ingredients, could still lead to a strong recommendation for the user. This enhances the system’s transparency by showing that the recommendation is not solely dependent on a single brand or fixed set of attributes but can adapt to user preferences, as long as the core characteristics of the product align with the user’s needs.

Counterfactual Analysis Results and Their Interpretation

For a given path to undergo counterfactual analysis, the counterfactual scenarios derived from validated neighbors are evaluated by the recommender system engine. The plausible attributes, according to the engine, are presented in four groups: brand, category, word, and related products. The revelations of each group of attributes are as follows:

- **Brand Influence:** If the recommended product had been associated with a different brand within the same category and still met the threshold score, this implies that brand substitutability exists within certain user preferences, indicating flexibility in brand choice.

- **Word Influence from Reviews:** Altering descriptive attributes (for example adding “animal cruelty free”, “organic”) and finding that the product still surpasses the score threshold suggests the potential descriptive qualities that the user will find favorable for the recommended product.
- **Connectivity and Exposure:** Introducing hypothetical connections to recommended product and products (e.g., often bought together or viewed together) and observing their impact on the recommendation score could highlight potential for cross-selling strategies.

4.1.3 Key Marketing Findings and Enhancing Marketing Strategies

This section outlines the marketing insights derived from our counterfactual analysis:

1. **Brand Associations:** The wide range of brands identified in the counterfactual analysis suggests that the recommended product has broad appeal across different market segments.
2. **Key Descriptors:** Words like "stays" and "eyelash" provide insights into the product's key selling points, which could be emphasized in marketing materials.
3. **Related Product Opportunities:** The identification of related products that influence recommendations presents cross-selling and bundling opportunities.

These insights can be utilized to enhance marketing strategies in several ways:

1. **Targeted Advertising:** By understanding the diverse attributes that lead to a recommendation, marketers can create more targeted and effective advertising campaigns.
2. **Product Positioning:** The variety of associated categories and brands allows for flexible product positioning to appeal to different customer segments.
3. **Content Marketing:** The identified key words and categories can inform content creation, ensuring that product descriptions and marketing materials align with the attributes that drive recommendations.
4. **Cross-Promotion:** The related products identified in the analysis can guide cross-promotion strategies, potentially increasing sales across product lines.
5. **Dynamic Pricing:** Understanding the range of brands associated with a product can inform pricing strategies, potentially allowing for price adjustments based on perceived brand value.
6. **Product Development:** Insights into the attributes that drive recommendations

can inform future product development, ensuring new products are designed with recommendation potential in mind.

By using these marketing insights derived from our counterfactual analysis, businesses can improve their recommendation systems and enhance their overall marketing effectiveness.

4.2 Ablation Study: Impact of Community Detection and Outlier Removal

In addition to the qualitative case study presented in Section 4.1, a further evaluation was conducted to assess the impact of the proposed filtering approach on counterfactual generation. Specifically, we performed an ablation study to compare runs with and without the key filtering steps—community detection and outlier removal. This section details the ablation settings and provides quantitative evidence that these filtering methods significantly reduce the search space while preserving most of the useful counterfactual scenarios.

4.2.1 Experimental Setup

The objective of the ablation study was to measure how community detection (CD) and outlier removal (OR) affect:

- The number of candidate attributes (words, brands, categories, related products) considered in counterfactual analysis,
- The number of final “eligible” counterfactuals that pass the scoring threshold (i.e., would remain recommended in plausible “what-if” scenarios), and
- The overall number of scenarios tested (which reflects computational complexity).

We considered four distinct configurations:

1. **Filtering = OFF:** No community detection and no outlier removal.
2. **Filtering = ON:** Both community detection and outlier removal enabled.
3. **Community_Filter = ON, Outlier_Filter = OFF:** Only community detection is applied.
4. **Community_Filter = OFF, Outlier_Filter = ON:** Only outlier removal is applied.

For each configuration, the system generated counterfactual scenarios for the same set

of user–product pairs. We measured:

- The count of “related” attributes detected for each type (categories, brands, words, related products),
- Their proportion relative to the total knowledge graph (e.g., $\text{categories_percentage} = \frac{\text{retained categories}}{\text{total possible categories}}$),
- The total number of counterfactual scenarios enumerated (`num_cf_scenarios`), and
- The number of “eligible” (plausible) scenarios that exceed the scoring threshold (`num_eligible_scenarios`).

4.2.2 Results Overview

Table 4.1 summarizes the ablation results for the first user discussed in Section 4.1, while Tables 4.2–4.5 present corresponding results for the four additional users.

Table 4.1: Ablation Study Results for User 0 (Community vs. Outlier Filtering).

Configuration	Categories	Words	Related Products	Counterfactuals
Filtering OFF	15,361	25,532	1,470,000	23
Filtering ON	7,500	12,000	382,000	21
Community ON, Outlier OFF	...	...	...	...
Community OFF, Outlier ON	...	...	...	...

4.2.3 Comparison Across Additional Users

To validate the consistency of these findings, we repeated the same four configurations for four other users. Tables 4.2–4.5 show comparable results:

Table 4.2: Results for User 1

Configuration	Categories	Words	Related Products	Counterfactuals
...	...	...	...	...

Table 4.3: Results for User 2

Configuration	Categories	Words	Related Products	Counterfactuals
...	...	...	...	...

Table 4.4: Results for User 3

Configuration	Categories	Words	Related Products	Counterfactuals
...	...	...	...	...

Table 4.5: Results for User 4

Configuration	Categories	Words	Related Products	Counterfactuals
...	...	...	...	...

4.2.4 Discussion

These ablation results suggest that community detection and outlier removal both play a crucial role in making counterfactual analysis more tractable while preserving the high-value explanations. By pruning commonly connected but uninformative nodes (outlier removal) and confining search to semantically relevant subgraphs (community detection), the method avoids exploring an exponential number of low-impact “what-if” scenarios. As shown in the tables, filtering often cuts the scenario space by over 70%, yet retains 90% or more of the ultimately eligible counterfactuals.

Such substantial computational savings is essential in practical deployments where real-time or near-real-time responsiveness is needed. Furthermore, the minimal impact on the final results indicates that the pruned attributes are largely redundant or uninformative from a user-modeling standpoint. This not only streamlines the analysis but also improves interpretability by limiting the number of potential counterfactual explanations that analysts or end-users must review.

4.3 Limitations and Challenges

1. **Computational Complexity:** The process of generating and evaluating counterfactual scenarios for each recommendation is computationally intensive, especially for large-scale knowledge graphs. This limits the real-time application of this method in production recommender systems.
2. **Attribute Independence Assumption:** Our analysis treats each attribute independently, which may not always reflect real-world scenarios where attributes can be correlated. This simplification might lead to some unrealistic counterfactual

scenarios.

3. **Threshold Sensitivity:** The choice of threshold score for determining relevant counterfactuals can significantly impact the results. Finding the optimal threshold that balances inclusivity and relevance remains a challenge.
4. **Temporal Aspects:** Our analysis doesn't account for temporal changes in user preferences or product popularity, which could affect the relevance of counterfactual scenarios over time.
5. **Bias Reinforcement:** There's a risk that focusing on attributes that lead to recommendations could reinforce existing biases in the recommender system.

Chapter 5

Conclusion

5.1 Summary of Key Findings

The implementation of counterfactual analysis on the path-based knowledge graph recommender system has yielded significant insights into the explainability of AI-driven recommendations. Our research demonstrates that by examining alternative scenarios and attribute modifications, we can shed light on the decision-making process of complex recommender systems. This approach allows us to move beyond the "black box" nature of many AI systems and provide clearer explanations for why specific products are recommended to users.

A key finding of our study is the identification of influential attributes and path types in the recommendation process. By analyzing different path structures, such as 'mentions' -> 'rev_described_by' and 'purchase' -> 'also_viewed' -> 'rev_also_bought', we have uncovered the underlying logic driving the recommender system's choices. This granular understanding of path influences enhances the system's transparency, allowing both developers and users to better comprehend the factors contributing to each recommendation.

Moreover, our counterfactual analysis revealed the relative importance of various product attributes, including brand associations, descriptive keywords, and product relationships. By systematically altering these attributes and observing the resulting changes in recommendation scores, we have developed a framework for explaining why certain products are recommended over others. This approach not only improves the explainability of the recommender system but also provides valuable insights into user preferences and decision-making patterns. However, it is important to note the limitations of our method,

such as the computational complexity and the assumption of attribute independence, which present opportunities for future research to further refine and expand upon this explainable AI approach in recommender systems.

5.2 Contributions to Explainable AI in Recommender Systems

Our methodology makes several key contributions to the field of explainable AI in recommender systems. First, it provides a novel approach to generating "what-if" scenarios, allowing for a more comprehensive understanding of the system's decision boundaries. This goes beyond traditional explanation methods that focus solely on the attributes present in the recommended item.

Second, our approach offers a more nuanced view of feature importance in recommendations. By identifying which attributes could lead to the same recommendation, we provide insights into the relative influence of different features, enhancing the interpretability of the system's decisions.

5.3 Marketing and Business Implications

From a marketing perspective, our analysis reveals valuable insights for product positioning and targeted advertising. The diverse range of brands and categories associated with recommended products suggests opportunities for flexible marketing strategies. Furthermore, the identification of key descriptive words and related products opens avenues for content marketing and cross-promotion strategies.

5.4 Future Research Directions

Future research in the domain of path-based knowledge graph recommender systems could explore the integration of scenarios that involve multiple counterfactual attributes simultaneously. This approach would address the complexity of real-world decision-making processes where multiple factors are often intertwined, affecting user preferences and product recommendations concurrently. By analyzing the combined effects of various attributes, researchers could gain a more nuanced understanding of their interdependencies and the synergistic effects on recommendation outputs. Additionally, dynamic

alterations of embeddings represent a promising area for enhancing the adaptiveness and accuracy of counterfactual explainability of recommender systems. Such efforts would advance the explainability of these systems, enabling clearer insights into how and why recommendations change under varied conditions.

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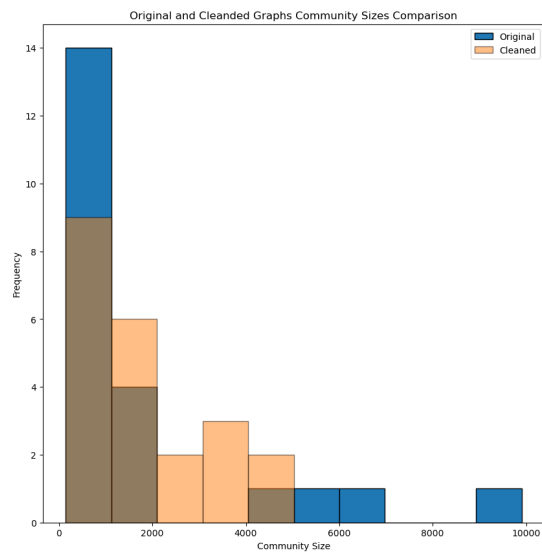
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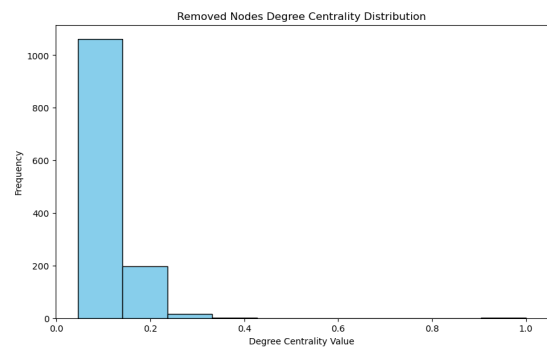
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Appendix

Various statistical methods were tested to detect outlier nodes. The impact of removing these nodes is illustrated in the plots from figure 5.1 to figure 5.5 on page 56 and table 5.1 on page 57 . below.

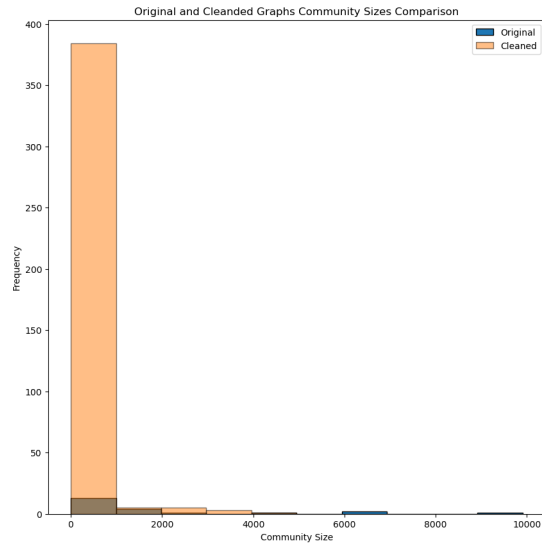


(a) Community Sizes Comparison (Standard Deviation)

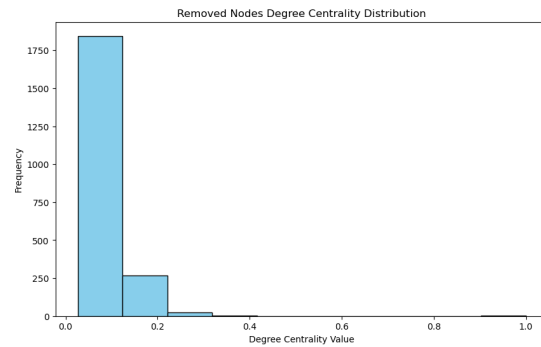


(b) Degree Centrality Distribution of Removed Nodes (Standard Deviation)

Figure 5.1: Comparison of Community Sizes and Degree Centrality Distribution of Removed Nodes Using the Standard Deviation Method

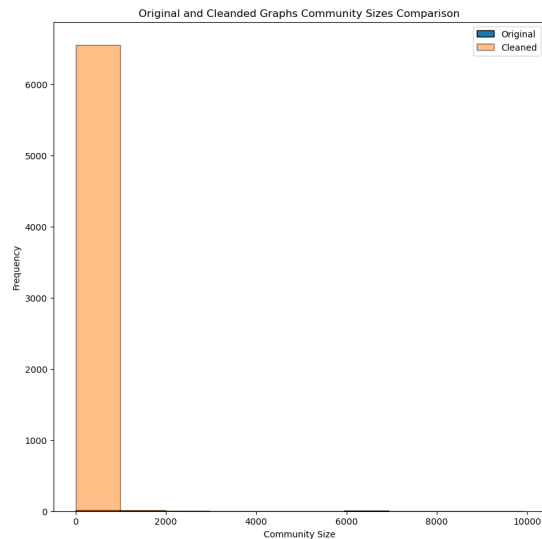


(a) Community Sizes Comparison (Percentile)

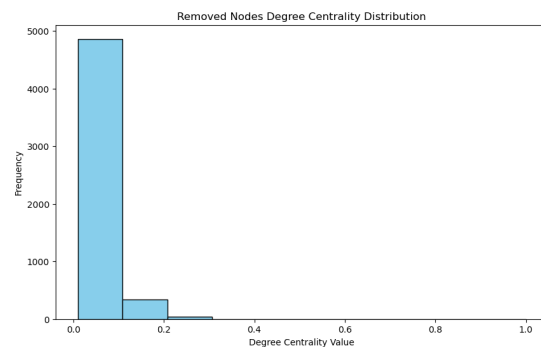


(b) Degree Centrality Distribution of Removed Nodes (Percentile)

Figure 5.2: Comparison of Community Sizes and Degree Centrality Distribution of Removed Nodes Using the Percentile Method

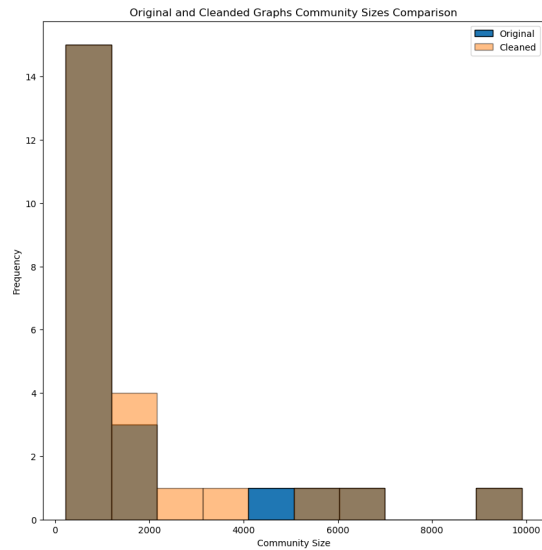


(a) Community Sizes Comparison (IQR)

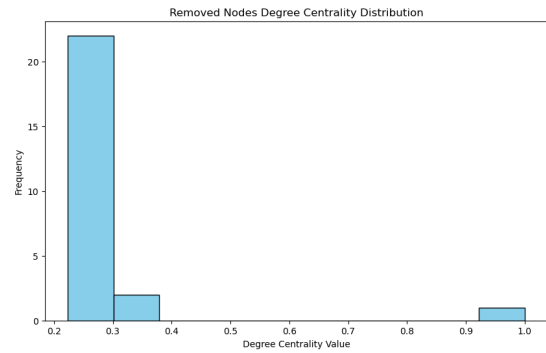


(b) Degree Centrality Distribution of Removed Nodes (IQR)

Figure 5.3: Comparison of Community Sizes and Degree Centrality Distribution of Removed Nodes Using the IQR Method

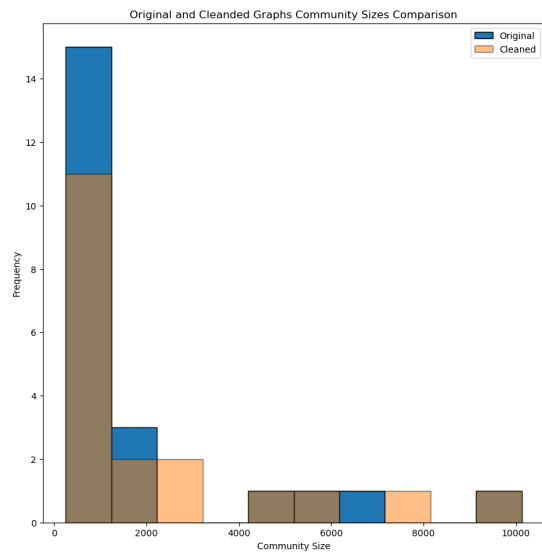


(a) Community Sizes Comparison (Knee Method)

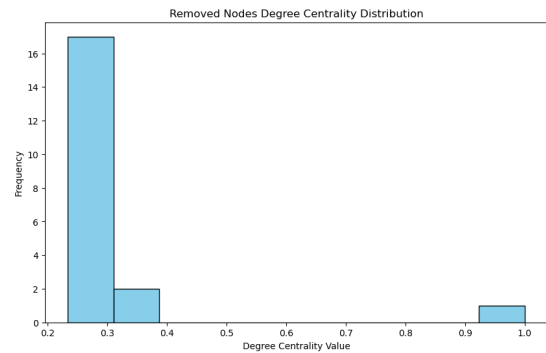


(b) Degree Centrality Distribution of Removed Nodes (Knee Method)

Figure 5.4: Comparison of Community Sizes and Degree Centrality Distribution of Removed Nodes Using the Knee Method



(a) Community Sizes Comparison (Top 20 Method)



(b) Degree Centrality Distribution of Removed Nodes (Top 20 Method)

Figure 5.5: Comparison of Community Sizes and Degree Centrality Distribution of Removed Nodes Using the Top 20 Method

Method/Metric	# of Removed Nodes	# Nodes in Cleaned Graph	# of Communities in Cleaned Graph
Standard Deviation	1279	41391	22
IQR	5241	37429	6559
Percentile	2137	40533	398
Knee Method	25	42645	24
Top 20	20	42650	19

Table 5.1: Comparison of Outlier Detection Methods